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Github Repository: https://github.com/RAshtr/Git_and_GitHub_Assignment

Implemented Workflows:

Task 1. SSH Authentication & Initial Branching

- Configured local SSH key authentication with GitHub.
- Initialized branch RAshttr, staged core project files, pushed to remote, and merged cleanly into main.

Task 1 commands:

```
# Initialize git and checkout user branch
```

```
git init
```

```
git checkout -b RAshttr
```

```
# Stage project structure and initial commit
```

```
git add .
```

```
git commit -m "feat: Initial commit with Flask project structure on  
branch RAshttr"
```

```
# Link remote and push branch
```

```
git remote add origin
```

```
git@github.com:RAshtr/Git_and_GitHub_Assignment.git
```

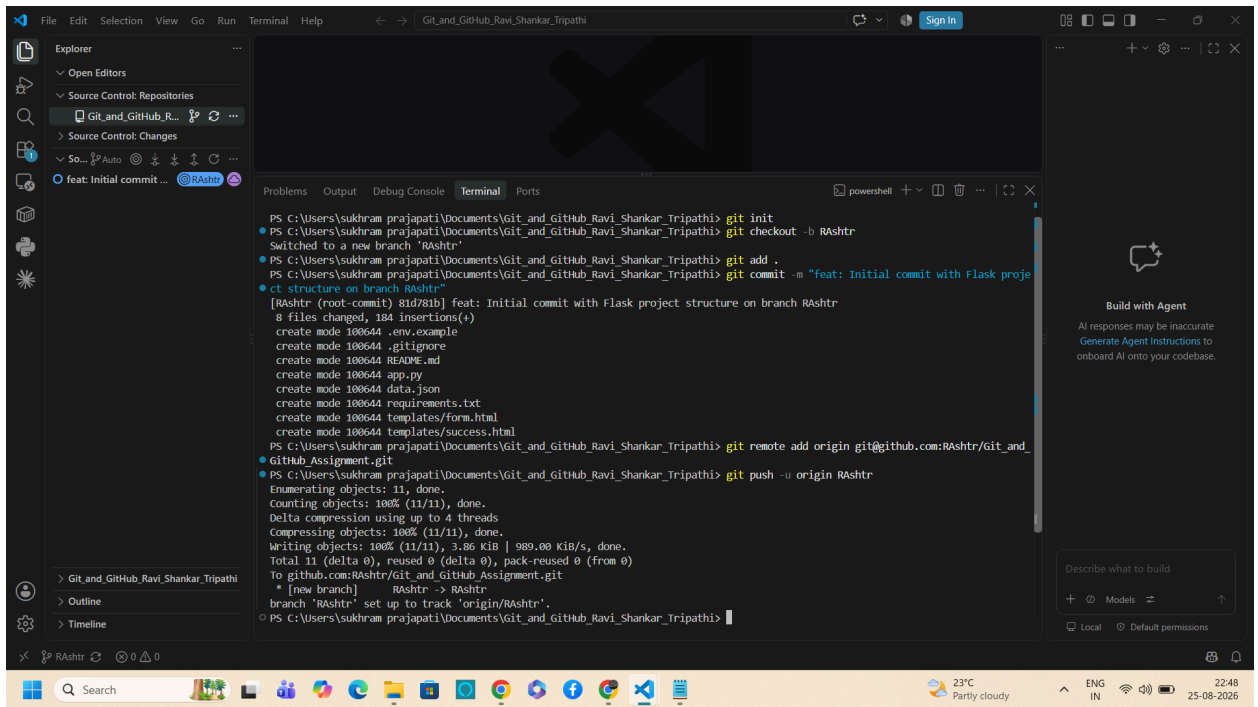
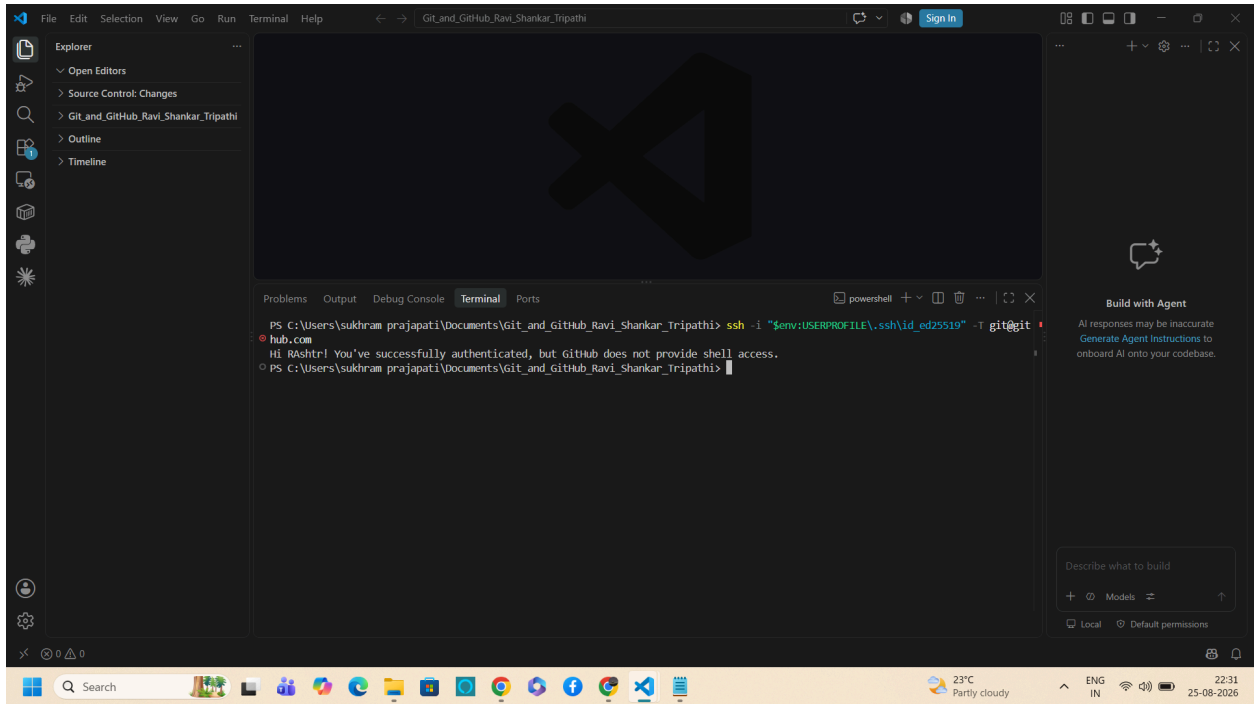
```
git push -u origin RAshttr
```

```
# Switch to main and perform initial merge
```

```
git checkout -b main
```

```
git merge RAshttr --no-ff -m "Merge branch 'RAshttr' into main"
```

```
git push -u origin main
```



Task 2: JSON Update & Merge Conflict Resolution

Simulated a real-world merge conflict on data.json across concurrent branches and resolved it manually.

Conflict Resolution Steps

1. Opened data.json in VS Code.
2. Selected Accept Incoming Change to prioritize updates from RAshttr_new.
3. Staged and committed the resolution:

Task 2 commands:

Branch creation and update

```
git checkout -b RAshttr_new
```

(Updated data.json content in feature branch)

```
git add data.json
```

```
git commit -m "Update data.json content in RAshttr_new branch"
```

Simulate conflicting update directly on main

```
git checkout main
```

(Modified identical lines in data.json on main)

```
git add data.json
```

```
git commit -m "Direct update on main branch to simulate conflict"
```

Trigger merge conflict

```
git merge RAshttr_new
```

Conflict Resolution Steps

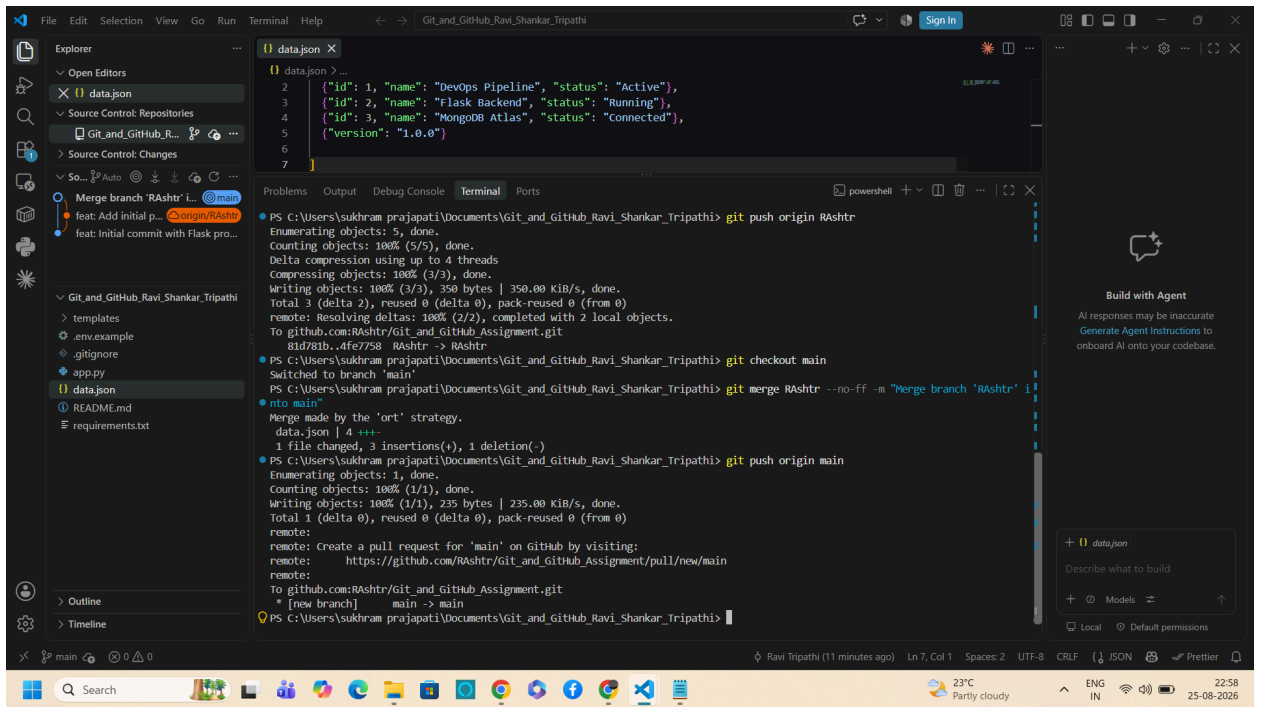
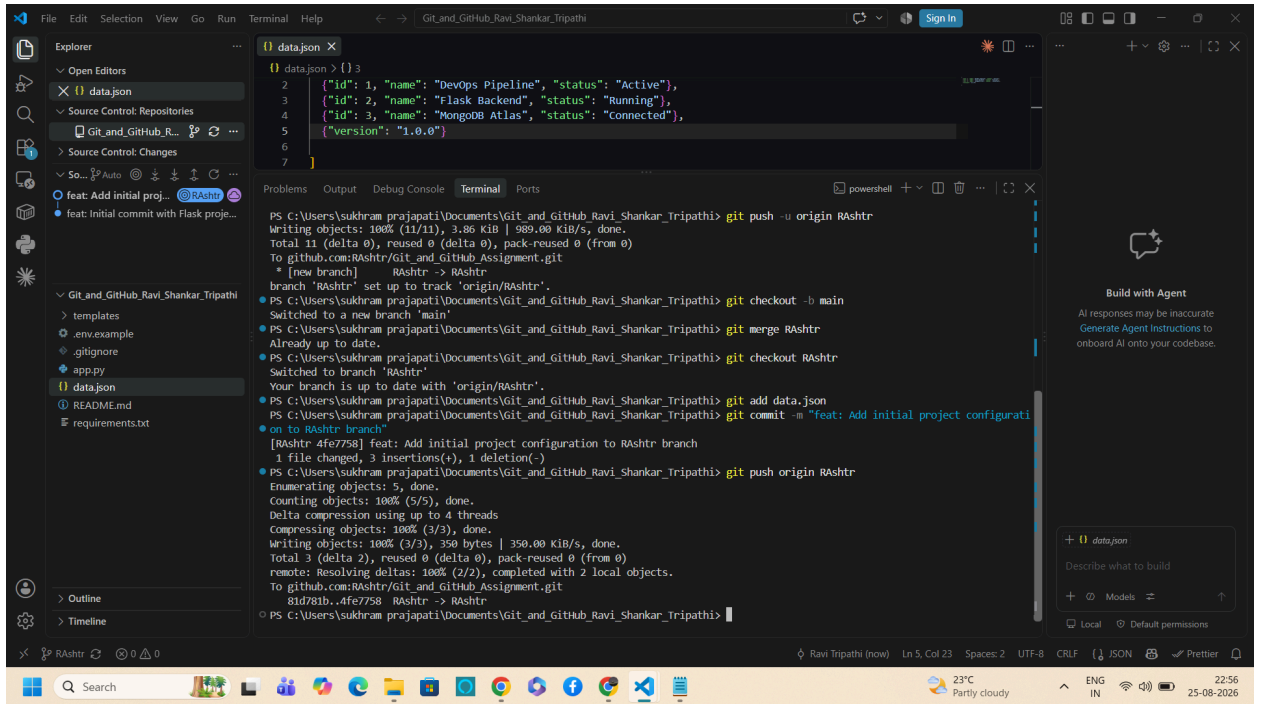
1. Opened data.json in VS Code.
2. Selected Accept Incoming Change to prioritize updates from RAshttr_new.
3. Staged and committed the resolution:

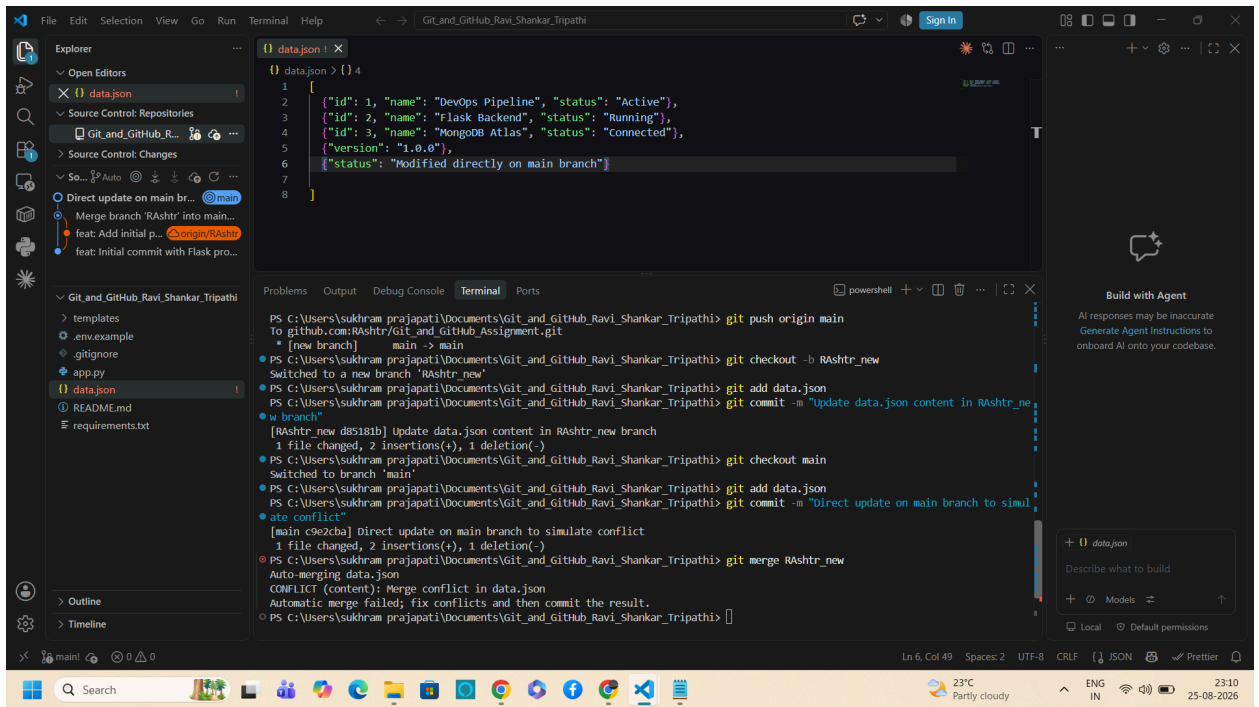
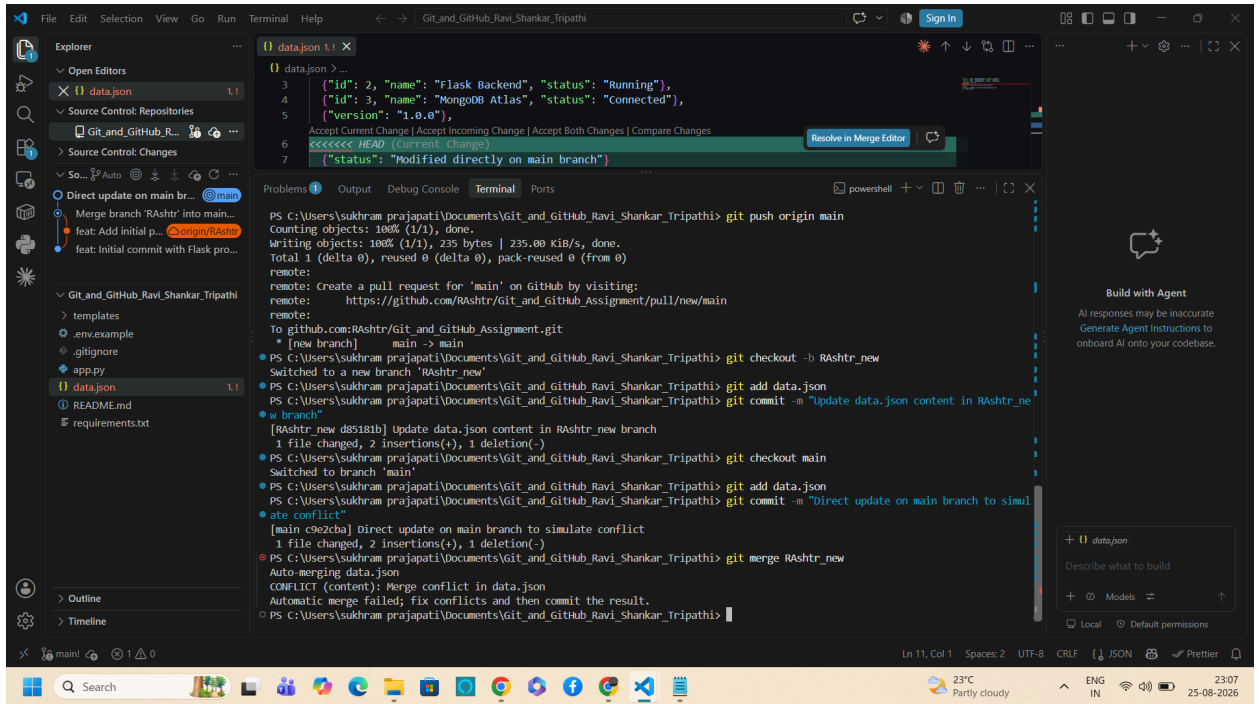
Commands:

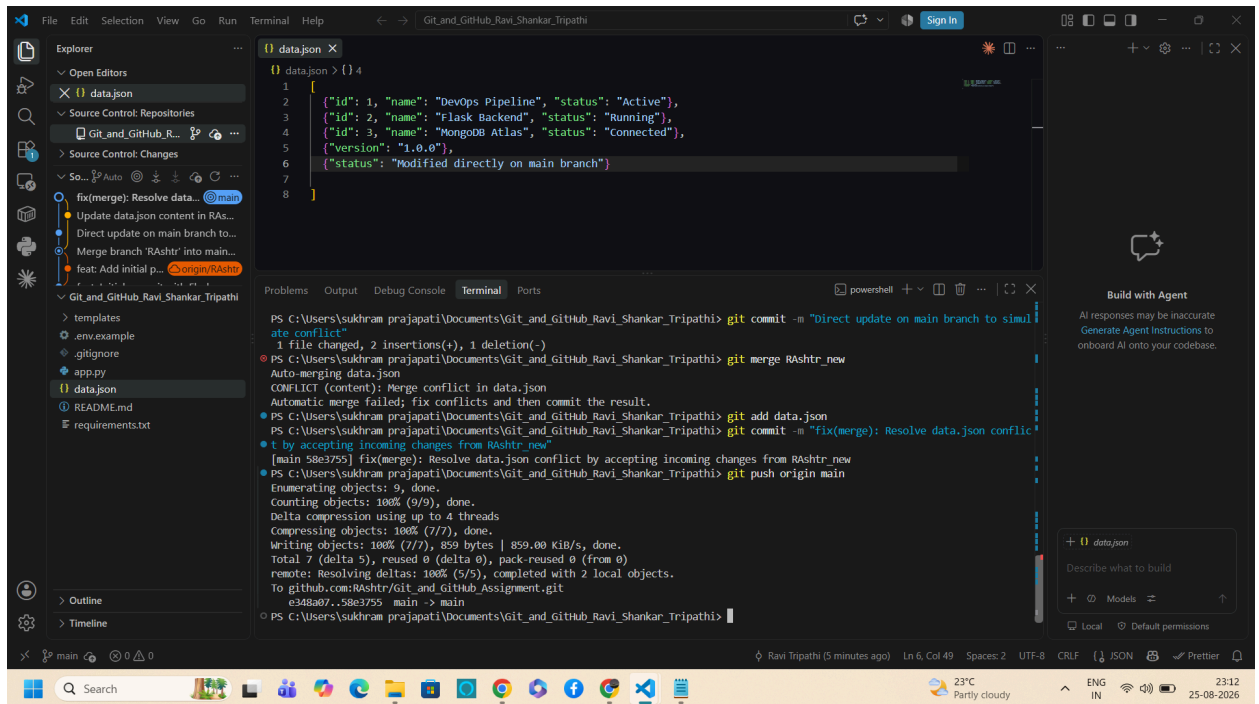
```
git add data.json
```

```
git commit -m "fix(merge): Resolve data.json conflict by accepting  
incoming changes from RAshttr_new"
```

```
git push origin main
```







Task 3. Parallel Feature Development:-

- **master_1 Branch:** Developed the frontend To-Do interface (templates/todo.html).
- **master_2 Branch:** Created the /submittodoitem backend API route in app.py for MongoDB Atlas data persistence.
- Merged both feature branches into main.

Task 3:

Feature A (Frontend UI in master_1)

Commands:

git checkout main

```
git checkout -b master_1
```

```
git add templates/todo.html
```

```
git commit -m "feat(frontend): Create To-Do form page with itemName  
and itemDescription"
```

```
git push -u origin master_1
```

Feature B (Backend API in master_2):

Commands:

```
git checkout main
```

```
git checkout -b master_2
```

```
git add app.py
```

```
git commit -m "feat(backend): Implement /submittodoitem route for  
MongoDB Atlas persistence"
```

```
git push -u origin master_2
```

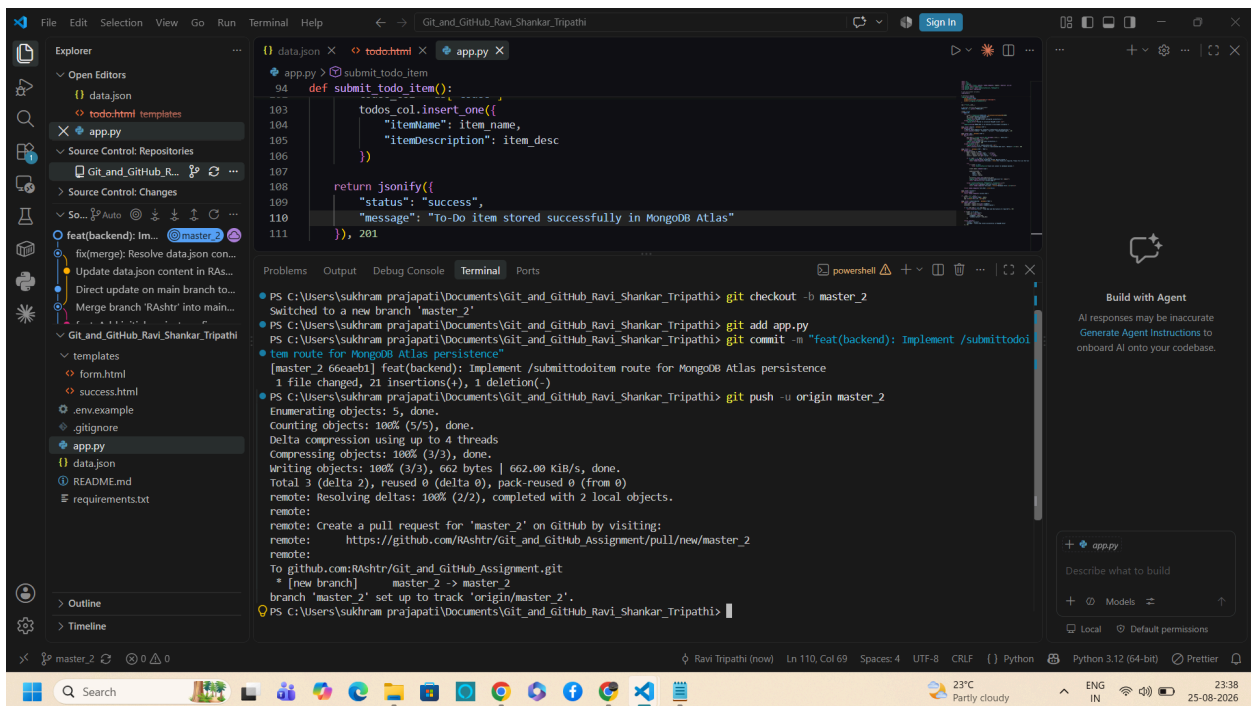
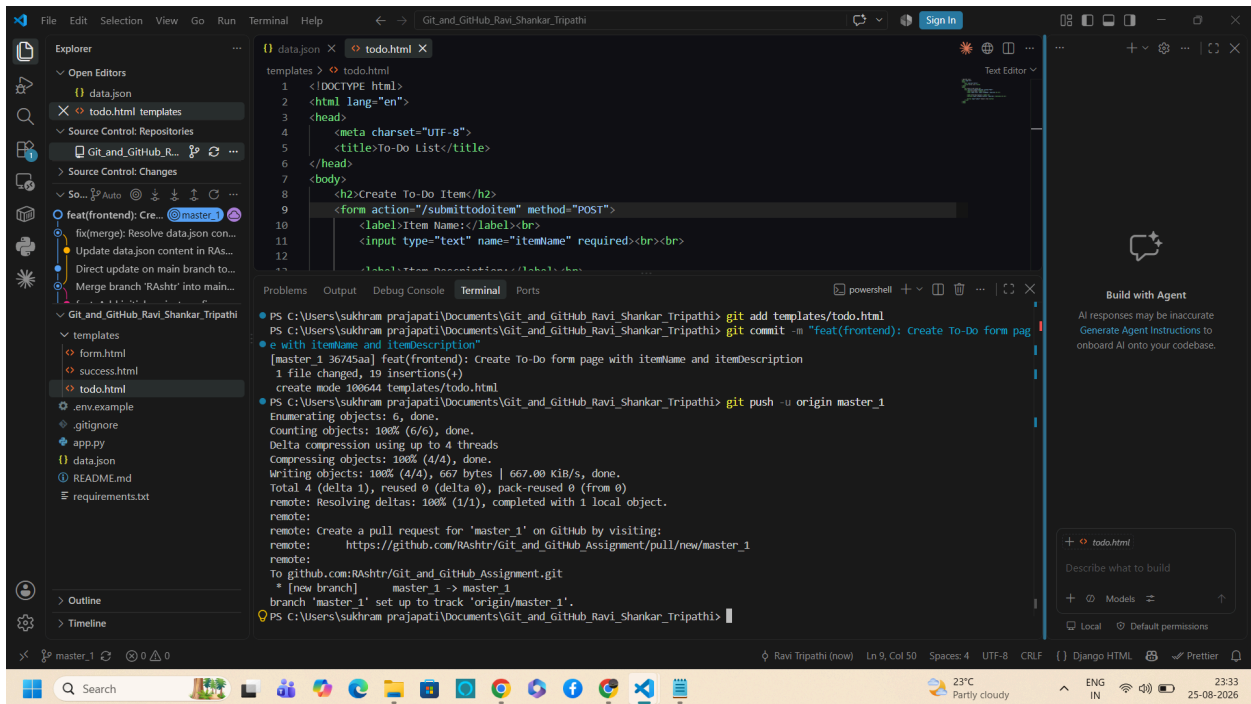
Integration into Main commands :

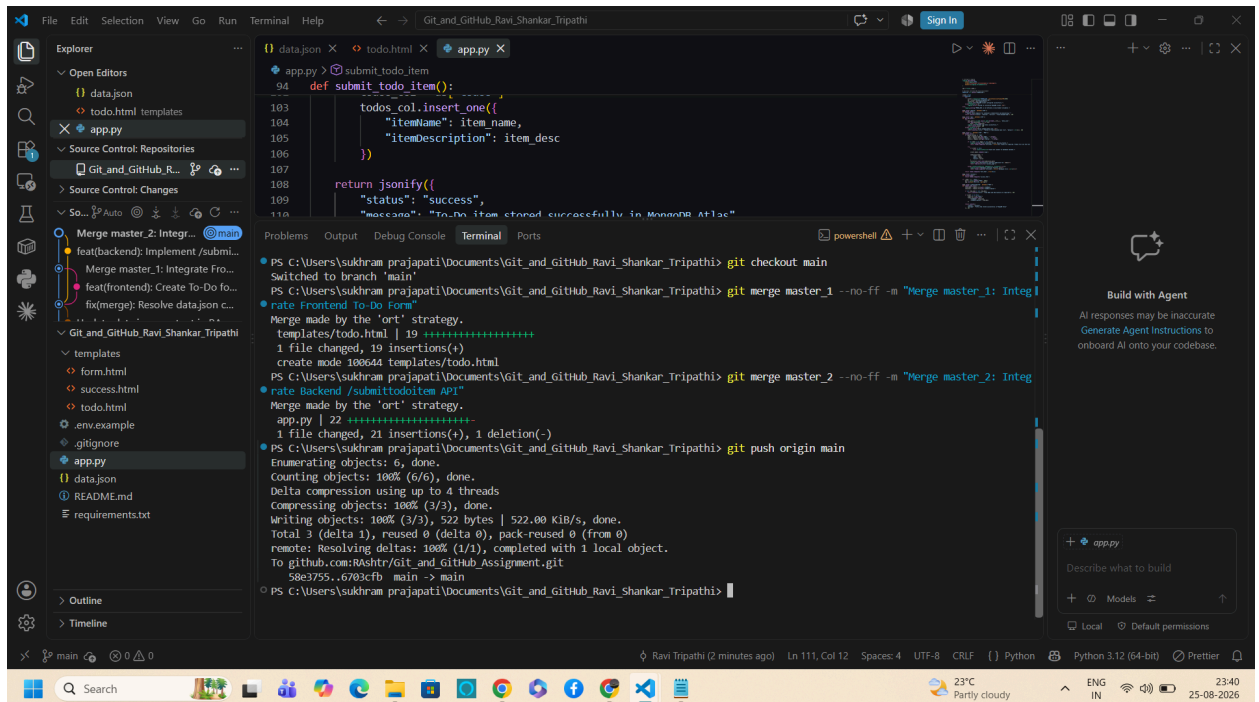
```
git checkout main
```

```
git merge master_1 --no-ff -m "Merge master_1: Integrate Frontend  
To-Do Form"
```

```
git merge master_2 --no-ff -m "Merge master_2: Integrate Backend  
/submittodoitem API"
```

```
git push origin main
```



Task 4. Sequential Commits, Soft Reset & Rebase:-

- Added Item ID, Item UUID, and Item Hash fields to `templates/todo.html` using three distinct sequential commits in `master_1`.
- Merged into `main` and executed `git reset --soft HEAD~2` to consolidate staged changes into a single clean commit.
- Rebased `main` onto `master_1` to preserve linear commit history

Task 4 commands:

Merge & Soft Reset

Merged `master_1` into `main`, then rolled back the top 2 commits using `git reset --soft` so that uncommitted changes remained staged.

```
git checkout main

git merge master_1

git reset --soft HEAD~2

git status

git commit -m "refactor: Consolidate UUID and Hash fields staged after
soft reset"
```

Git Rebase & Linear History :

```
git checkout master_1

git rebase main

# (Resolved in-flight template merge markers and continued)

git add templates/todo.html

git rebase --continue

# Verify commit graph

git log --oneline --graph -n 8

# Synchronize remote

git push origin main --force
```

git push origin master_1 --force

The screenshot shows the Visual Studio Code interface with the following details:

- Explorer:** The file tree on the left shows the project structure, including `templates` and `data.json`.
- Source Control:** The 'Source Control: Changes' view shows a commit titled 'feat: Add Item ID field to To-Do form'.
- Terminal:** The terminal window at the bottom displays the execution of the following commands:

```
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git checkout master_1
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git add templates/todo.html
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git commit -m "Feat: Add Item ID field to To-Do form"
[master 1 6cb1744] feat: Add Item ID field to To-Do form
1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git add templates/todo.html
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git commit -m "Feat: Add Item UUID field to To-Do form"
[master 1 7307046] feat: Add Item UUID field to To-Do form
1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git add templates/todo.html
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git commit -m "Feat: Add Item Hash field to To-Do form"
[master 1 f643263] feat: Add Item Hash field to To-Do form
1 file changed, 2 insertions(+), 1 deletion(-)
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git push origin master_1
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 4 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.01 KiB | 38.00 KiB/s, done.
Total 12 (delta 9), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (9/9), completed with 3 local objects.
To github.com:RaviShankarT/Assignment.git
36745aa..f643263 master_1 -> master_1
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi>
```
- Editor:** The `todo.html` file is open, showing HTML code for a form with fields for item name, item ID, and item hash.
- Right Panel:** The 'Build with Agent' panel is visible, showing a message about AI responses and a button to generate agent instructions.

The screenshot shows the Visual Studio Code interface with the following details:

- Source Control:** The 'Source Control: Changes' view shows a commit titled 'refactor: Consolidate ...'.
- Terminal:** The terminal window at the bottom displays the execution of the following commands:

```
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git rebase main
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 7307046... # feat: Add Item UUID field to To-Do form
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git add templates/todo.html
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git rebase --continue
dropping f643360ec32d2979e7b1e1c6bdf0c54a67869 feat: Add Item Hash field to To-Do form
Successfully rebased and updated refs/heads/master_1.
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi> git log --oneline --graph -n 8
* f70bad8 (HEAD -> master_1, main) refactor: Consolidate UUID and Hash fields staged after soft reset
* 54325b5 Merge master_1: Integrate Frontend To-Do Form
* 36745aa feat(frontend): Create To-Do form page with itemName and itemDescription
* 58e3755 fix(merge): Resolve data.json conflict by accepting incoming changes from Rashttr_new
* d85181b (Rashttr_new) Update data.json content in Rashttr_new branch
* c9e2cba Direct update on main branch to simulate conflict
* e348a07 Merge branch 'Rashttr' into main
* 4fe7758 (origin/Rashttr, Rashttr) feat: Add initial project configuration to Rashttr branch
PS C:\Users\sukhram prajapati\Documents\Git_and_Github_Ravi_Shankar_Tripathi>
```
- Editor:** The `todo.html` file is open, showing HTML code for a form with fields for item name, item ID, and item hash.
- Right Panel:** The 'Build with Agent' panel is visible, showing a message about AI responses and a button to generate agent instructions.

